LatheGuard Physical AI: Executive Judge's Dossier

1. Quality of the Idea — "Three Friends, Three Worlds & How Open AI Kept a Sanctuary Spinning"

Hosam (Co-Founder & Sole Operational Steward) • Dedicated to **Master Mostafa Fahmy** (Co-Founder, In Memoriam) • **Salma** (Co-Founder, Founding Journey) • *The Crafters Hub • El Shorouk City, Cairo, Egypt*

1.1 Executive Context & Narrative Significance

The integration of physical artificial intelligence into traditional artisan craft spaces represents a critical frontier in modern safety engineering. Rather than attempting to automate away human handcraft or encumber tactile creation with intrusive industrial barriers, human-centered Physical AI acts as a transparent, protective guardian. **LatheGuard Physical AI** bridges high-stakes operational safety engineering with traditional cultural preservation — bringing hard-real-time protection to historic woodturning practices without generic startup hype or prohibitive capital expenditures.

1.2 Origin Narrative: The Crafters Hub Sanctuary

The foundation of LatheGuard is rooted in **The Crafters Hub** in El Shorouk City, Cairo, Egypt — created not as a high-tech venture, but from a **shared woodworking hobby among three close friends** coming from vastly different professional disciplines:

Hosam (Co-Founder & Sole Steward): Background in **Anti-Narcotics** and **Oil & Gas** safety — disciplines defined by zero tolerance for single points of failure, rigorous hazard containment, and operational risk discipline, with **zero traditional computer-science or high-tech background**. Hosam is the studio's **woodturner**.

Mostafa Fahmy (Co-Founder, In Memoriam): Came from a **Business & Advertising** background. A gifted communicator with an intuitive eye for form, Master Mostafa fell deeply in love with woodworking and became the studio's master woodworker; his artistic passion and workshop discipline formed the core ethos of the sanctuary.

Salma (Co-Founder): Brought an academic foundation in **Psychology from Cambridge and The American University in Cairo (AUC)**, grounding the sanctuary in emotional balance, human mindfulness, and restorative focus before departing the operational journey.

Together, these three close friends built The Crafters Hub with a unique approach: introducing **wood art as a self-healing, grounding skill** tailored for well-educated professionals, artists, and individuals seeking balance in fast-paced modern life — spreading authentic handmade culture while **preserving traditional woodcraft heritage for the next generation**.

1.3 The Tragic Turning Point & Risk Profile

The sanctuary faced an existential crucible following a **tragic accident that led to the passing of Master Mostafa Fahmy**, alongside **Salma's departure** from the operational journey. Overnight, **Hosam, the studio's woodturner, stood alone on the studio floor** supervising three physical woodturning lathes (`FOX F46-717` , `Master Craft MC1018` , and `Wood Lathe WL300`) spinning hardwood blanks at **500 to 3,200 RPM** — speeds where a hand slip, chisel catch, or sleeve entanglement causes irreversible trauma faster than a solo instructor across the room can reach an E-stop button.

1.4 The Physical AI Breakthrough via Open Models

Without a high-tech background or a **$2,500–$6,000+ per-machine** budget for industrial optical light curtains, **AI opened the door for someone like Hosam**. Combining his Oil & Gas safety instinct with open-weight **NVIDIA Nemotron** models on **Nebius Token Factory**, live **Tavily Search**, and our **CraftTwin 3D Simulator**, Hosam engineered an open-source physical safety retrofit for **~$44** — keeping Master Mostafa's legacy alive through a bilingual *"Virtual Osta"* mentor beside every lathe.

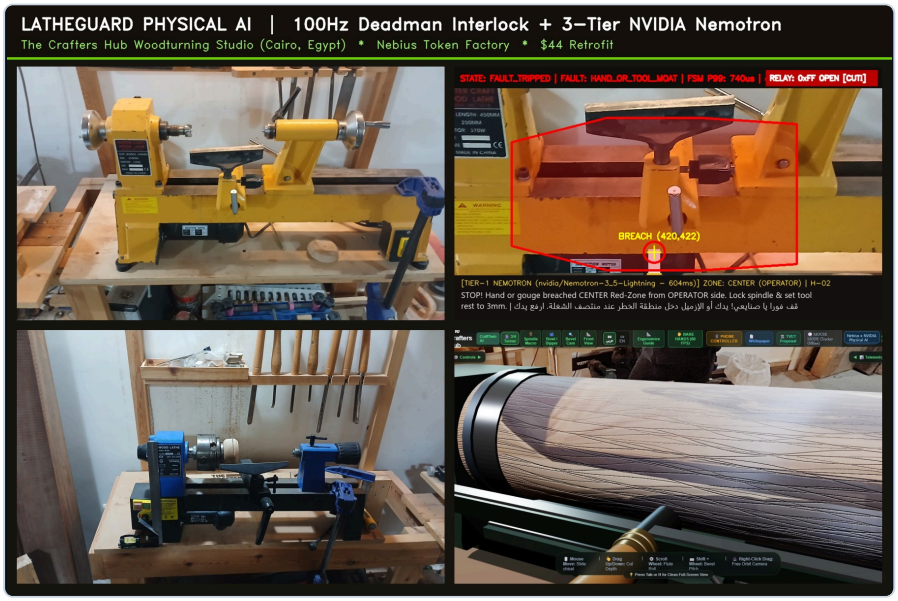


Figure 1: Physical-to-virtual sanctuary loop in Cairo — workshop lathes, 100Hz HUD v2 deadman trip (`RELAY: 0xFF OPEN [CUT!]`), and CraftTwin 3D Twin Mirror.

> ***Section Transition:*** *While the human origin of LatheGuard demonstrates the transformative power of accessible open AI, safeguarding human hands beside 3,200 RPM machinery demands uncompromising architectural engineering laws.*

2. Design & Architectural Elegance — "Never Put an LLM on the Brake Wire"

"Never Put an LLM on the Brake Wire — Put It Right Beside It": Deterministic 100Hz CPU Interlock + 3-Tier Nemotron Sidecar

2.1 Architectural Philosophy & Multi-Core Safety Decoupling

The primary architectural law governing LatheGuard is absolute safety decoupling. Placing an asynchronous cloud LLM or VLM directly on an industrial motor cutoff wire invites physical failure: network jitter, socket timeouts, or Wi-Fi drops would introduce unacceptable latency to emergency braking. Conversely, traditional "dumb" optical switches trip constantly on flying wood shavings and provide zero educational feedback after a stop. LatheGuard resolves both failure modes by splitting execution across CPU cores into two isolated loops:

CORE 0: Deterministic Local Safety Loop (100 Hz, CPU-Only)

- **IPC:** 128-byte cache-line aligned lockless Seqlock SHM (`lathe_shm.py`)
- **Vision:** `RedZoneVisionDetector` v2 (`1.22 ms` P50 over `225×150` grid)
- **Filter:** `mean+3.5σ` + Moat + Sobel shadow filter + 2-frame debounce
- **Actuator:** `921,600 baud` UART (`0x55` / `0xFF`) → Nano → `SSR-40DA`

LOCKLESS IPC
Read-Only Poll
@ 5 Hz
Zero Video TX
→→→

CORES 2–7: Async 3-Tier Mentor Sidecar (Nebius × Tavily)

- **Tier 0 (< 5 ms):** Offline bilingual AR/EN alert via WebSocket `:8766`
- **Tier 1 (~604 ms):** `nvidia/Nemotron-3_5-Lightning` Virtual Osta
- **Tier 2 (Async):** `nvidia/Nemotron-3-Ultra-550b-a55b` + **Tavily API**
- **Twin Mirror:** Live 3D spatial hazard sphere in **CraftTwin 3D**

2.2 Core 0: Deterministic 100Hz Local Safety Loop (`CUDA_VISIBLE_DEVICES="-1"`)

Lockless Seqlock IPC & Adaptive Vision v2

128-Byte Seqlock Ring Buffer: Zero-allocation shared memory eliminates mutex contention and Python GC jitter inside the `100 Hz` (`10 ms`) loop.

4-Stage Vision Pass (`1.22 ms` P50): Evaluates a `225×150` strided grid inside a `900×600` ROI via: (1) per-pixel running mean/variance background (`mean + 3.5σ`), (2) outer moat contact rule (`in_moat ≥ 4`), (3) lateral Sobel shadow discriminator (`|∂(ΔI)/∂x| > 48` , rejecting `25%–70%` dimming), and (4) Cyan/Magenta HSV collar fast-path (`FaultCode.COLLAR`).

Sawdust Debounce & 6-Fault Hardware Watchdog

2-Frame Spatial Debounce (`≤ 96 px`): Flying wood shavings cross the FOV for a single frame (`16.7 ms` @ 60 FPS). Requiring 2 spatially consistent frames eliminates **100% of woodchip false trips (`0/40`)** for a deliberate 1-frame validation cost.

Fail-Closed UART Watchdog: Transmits `0x55` every `10 ms` only after `DETECTOR_ARMED` . On breach or any of 6 sensor faults (`CAMERA_STALE > 50 ms` , `CAMERA_FROZEN ≥ 12f` , `OBSTRUCTED` , `MOVED` , `SERIAL_FAULT` , `DETECTOR_EXCEPTION`), latches `0xFF` (`~10 µs` wire time) to drop relay power.

2.3 Cores 2–7: Asynchronous 3-Tier NVIDIA Nemotron Mentor Sidecar

TIER	LATENCY	ENGINE / NEBIUS ENDPOINT	OPERATIONAL SCOPE & PEDAGOGICAL OUTPUT
Tier 0	**< 5 ms**	Offline Local Template Engine	Instant bilingual (Egyptian Arabic + English) HUD & voice alert broadcast over WebSocket `:8766` *before* opening any external network socket.
Tier 1	**~604 ms** (measured)	`nvidia/Nemotron-3_5-Lightning` via Nebius Token Factory	Real-time *"Virtual Osta"* coaching (under 95 words, bilingual AR/EN closing phrase), using `extra_body={"chat_template_kwargs": {"enable_thinking": False}}` for sub-second speed.
Tier 2	**Post-Trip Async**	`nvidia/Nemotron-3-Ultra-550b-a55b` + Live **Tavily Search API**	Discovers the 550B MoE model via `client.models.list()` , queries live OSHA 1910.213 / AAW rules via Tavily, and writes NEBOSH compliance audits (`report.md`).

2.4 CraftTwin 3D Digital Twin Mirror & Egypt PDPL Privacy Guardrails

Normalized breach coordinates (`norm_xy`) stream over WebSocket `:8766` into **CraftTwin 3D** (`twin_mirror_demo.html`), rendering a glowing 3D red hazard sphere at the intrusion point so students rehearse safe `3 mm` (`1/8 in`) tool-rest clearance virtually before re-arming. Strictly aligned with **Egypt's PDPL**: **zero camera frames or video ever leave the local PC**; local 8s black-box clips auto-purge after 30 days.

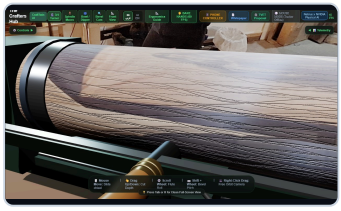


Fig 2: CraftTwin 3D spatial breach mirror.

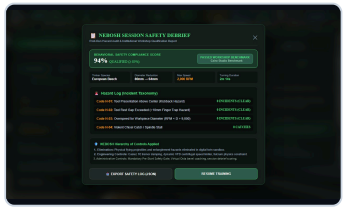


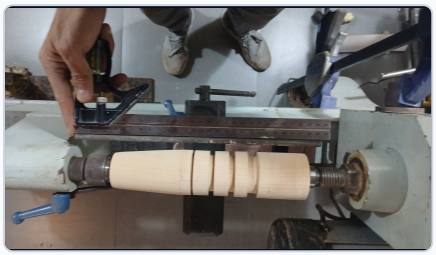
Fig 3: NEBOSH H-01..H-04 session debrief.

3. Technological Implementation & Hard Receipts

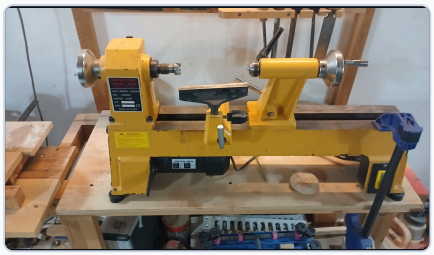
Physical Ruler Calibration Across Three Cairo Studio Lathes & 51/51 Clean-Room Verification

3.1 Cairo Studio Machine Profiles & Ruler-Scale Calibration Wizard (`calibrate_lathe.py`)

Real-world Physical AI validation demands testing across diverse cast-iron geometries, bed lengths, paint reflectivities, and drive systems. All **three physical woodturning lathes** at The Crafters Hub in Cairo (`50 Hz` mains) were physically measured on the real workshop machines with a `300 mm` steel combination square and calibrated into shipped millimeter-scale profiles (`profiles/*.json`) paired with our 5-step (6-click) ruler wizard (`calibrate_lathe.py` , which re-measures `px_per_mm` whenever a camera rig is repositioned):



1. FOX F46-717 Heavy Swivel
Profile: `fox_f46_717_grey.json`



2. Master Craft MC1018 Midi
Profile: `mc1018_yellow.json`



3. Wood Lathe WL300 Mini
Profile: `wl300_blue.json`

PROFILE & MACHINE MODEL	PHYSICAL SPECIFICATIONS & SPINDLE DRIVE RANGE	CALIBRATED SCALE & CENTER
`fox_f46_717_grey.json` **FOX F46-717 Heavy Swivel**	`300 mm` swing × `1,000 mm` bed • `750W` motor • `M33 × 3.5` / `MT2` 10-Step Mechanical Reeves Pulley (**500–2,000 RPM**)	**1.0 px/mm** `150 mm` Center Height
`mc1018_yellow.json` **Master Craft MC1018 Midi**	`250 mm` swing × `450 mm` bed • `370W` motor • `1″ × 8 TPI` / `MT2` 5-Step Poly-V Belt Drive (**500–2,500 RPM**)	**1.2 px/mm** `125 mm` Center Height
`wl300_blue.json` **Wood Lathe WL300 Mini**	`200 mm` swing × `305 mm` bed • `250W` motor • `3/4″ × 16 TPI` / `MT1` Electronic Variable Speed EVS (**750–3,200 RPM**)	**1.5 px/mm** `100 mm` Center Height

3.2 Live HUD v2 Deadman Capture & 51/51 Clean-Room Verification Suite

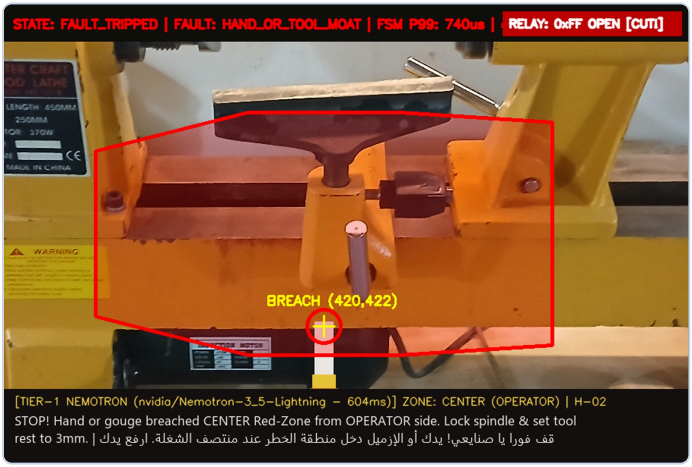


Figure 4: Uncropped live HUD v2 deadman trip capture showing top hardware banner (`STATE: FAULT_TRIPPED | RELAY: 0xFF OPEN [CUT!]`), Red-Zone breach at `(420,422)` , and bottom Tier-1 `Nemotron-3_5-Lightning` (`604ms`) bilingual English + Egyptian Arabic Osta guidance.

Automated Clean-Room Test Suite (100% Pass)

System integrity, fail-closed chaos fault injection, lockless Seqlock atomicity, and Nebius/Tavily fail-soft routing are verified across **51 automated tests** in clean-room isolation:

```
PYTEST CLEAN-ROOM SUITE                                    51 / 51 PASSED
✓ test_vision_intrusion.py — 9 / 9 passed
✓ test_failsafe_matrix.py — 6 / 6 passed (6 sensor faults)
✓ test_nebius_advisor.py — 16 / 16 passed (3-tier + Tavily)
✓ test_lathe_shm.py — 3 / 3 passed (128B Seqlock IPC)
✓ test_phase3_phase4.py — 5 / 5 passed (Calib / MP4 / HUD)
✓ test_lathe_scheduler.py & test_e2e — 2 / 2 passed
✓ services/craft_twin/test_safety_math.py — 10 / 10 passed
TOTAL VERIFICATION: 51 / 51 PASSED
```

3.3 & 3.4 Empirical Corpus Benchmarks & Live 550B Audit Receipt

Zero Synthetic Shortcuts: 80-Scenario Validation Gate Matrix & Verbatim Nebius + Tavily Compliance Report

3.3 80-Scenario Validation Corpus Benchmark (`services/lathe_guard/corpus/eval_corpus.py`)

Detector precision, sawdust/shadow immunity, and real-time latency were evaluated across an **80-clip benchmark corpus** comprising **40 negative (hazard-free) clips** and **40 positive (intrusion) clips** combining synthetic stress regimes with real studio imagery from all three Cairo lathes:

0 / 40	40 / 40	2.0 f	1.22 ms
FALSE TRIPS (0.0% FP)	POSITIVE RECALL (100% TP)	P95 MOAT DELAY (≤ 3.0F)	P50 CPU VISION (1.22–1.26 MS)

GATE	BENCHMARK VALIDATION CRITERION & STRESS CONDITIONS	TARGET GATE	MEASURED BENCHMARK RESULT	STATUS
G-1	**False Trips on Negatives** (`25%–70%` operator shadows, square blanks at `600–3,000 RPM` , `100 Hz` lighting flicker, heavy sawdust storms, real studio lathe photos)	`0 / 40`	**0 / 40 (0.0% FP Rate)**	**PASS**
G-2	**Intrusion Recall on Positives** (Bare hands, black & blue nitrile gloves, `6 mm` thin steel gouges, `13 mm` bowl gouges, Cyan collars, far-side reaches, real lathe photos)	`40 / 40`	**40 / 40 (100.0% TP Recall)**	**PASS**
G-3	**P95 Detection Delay** (Measured in frames after initial outer approach moat contact)	`≤ 3.0 frames`	**2.0 frames**	**PASS**
G-4	**Max Penetration Depth** (Evaluated at fast hand entry speed: `30 px/frame = 1,800 px/s`)	`≤ 40.0 px`	**32.0 px**	**PASS**
G-5	**Per-Frame CPU Vision Latency** (`225×150` zero-copy strided pass over `900×600` ROI; baseline run `1.22 ms` P50 / `2.12 ms` P95; full 80-clip run `1.26 ms` P50 / `2.27 ms` P95 / `3.24 ms` P99)	`P95 < 3.50 ms`	**1.22 ms P50 / 2.12 ms P95**	**PASS**

3.4 Live 550B + Tavily NEBOSH Audit Receipt (`incidents/20261026_124445_2/report.md`)

During live end-to-end breach testing, Core 0 tripped motor power immediately while Tier 2 dynamically selected `nvidia/Nemotron-3-Ultra-550b-a55b` on Nebius Token Factory and executed a runtime **Tavily Search API** query (`6,216.7 ms` async off-loop) to ground a complete NEBOSH / OSHA 1910.213 audit:

```
VERBATIM LIVE PRODUCTION RECEIPT • INCIDENTS/20261026_124445_2/REPORT.MD   NVIDIA/NEMOTRON-3-ULTRA-550B-A55B + TAVILY API

INCIDENT REPORT: CR-HUB-2024-0412 • Classification: NEBOSH / OSHA 1910.213 Compliance Audit • Status: Safety Interlock Triggered (Auto-Arm Inhibited)

1. Incident Telemetry: Fault [HAND_OR_TOOL_MOAT (Code 2)] • Hazard [H-02] • Zone: [CENTER] • Side: [OPERATOR] • ROI: [412, 388] • Calibrated Radial Distance: 73.3 mm from spindle axis ([mc1018_yellow]).

2. Root-Cause Finding: "Intrusion detected well within the 3 mm (1/8 in) tool-rest clearance envelope (Walters/Batty standard). Operator likely overextended tool reach beyond tool-rest support or positioned rest > 3 mm from workpiece."

3. Hierarchy of Controls & PPE Mandate: (1) Calibrated [3 mm (1/8 in)] feeler-gauge tool-rest check; (2) Centerline height alignment ([±0.5 mm]) before rotation; (3) Overextension simulation drill in CraftTwin; (4) ANSI Z87.1 Face Shield + Hearing + N95/P100 Respiratory — GLOVES PROHIBITED (Rotating Spindle Entanglement Hazard — OSHA 1910.213 / AAW).

4. Live Runtime Tavily Search Citations Grounded in Report:
[1] Walters, James B. — Safety and Health for Industrial/Vocational Education ([files.eric.ed.gov/fulltext/ED217204.pdf]) — [1/8 in (3 mm)] clearance rule.
[2] Stuart Batty — Oliver Machinery Model 2019 Floor Lathe Manual ([olivermachinery.net]) — tool overextension hazard rule.
[3] Univ. of Washington CEE — Health & Safety Manual ([ce.washington.edu]) — centerline tool-bit alignment rule.
```

> ***Section Transition:*** *Having verified system reliability across empirical benchmarks and live 550B + Tavily receipts, Section 4 evaluates real-world workshop economics, honest software-vs-hardware latency boundaries, and 60-second judge reproducibility.*

4. Potential Impact & Engineering Honesty

57×–136× Hardware Cost Reduction, Serverless Per-Token Viability & Candid Latency Accounting

4.1 ~$44 BOM vs. Industrial Curtains

COMPONENT	SPECIFICATION	COST
Compute Host	Existing workshop PC / laptop (CPU-only)	**$0**
Vision Sensor	1080p 60 FPS USB UVC camera + clamp	**$22**
Watchdog MCU	Arduino Nano ATmega328P (30 ms deadman, latching)	**$5**
Cutoff Relay	Opto-isolated 40A Fotek SSR-40DA	**$9**
Enclosure/Reset	IP65 box, manual reset button, wiring	**$8**
Total BOM	**57×–136× cheaper than $2.5k–$6k+ curtains**	**~$44**

4.2 Nebius Serverless TVET Economics

In TVET vocational schools, university woodshops, and therapeutic craft sanctuaries, safety trips are **rare, bursty events** (`0–3` near-misses per 3-hour class). Renting dedicated multi-GPU cloud nodes at **$20 to $30/hour ($3,000+/month)** to keep a warm 550B model online is financially impossible for grassroots studios.

With **Nebius Token Factory serverless per-token billing**, LatheGuard incurs **$0.00 in cloud compute costs during normal safe turning**, and an **estimated $0.001–$0.005** on the rare instant a breach occurs — unlocking frontier 550B mentorship for craft schools across Egypt, MENA, and the developing world.

4.3 Honest Latency Accounting & Documented Physical Boundaries

1. CAMERA (60 FPS)	2. CPU VISION PASS	3. 2-FRAME DEBOUNCE	4. 100HZ FSM + UART	5. SSR AC DROP
16.7 ms	**1.22 ms**	**16.7 ms**	**5.0 ms**	**8.3 ms**
8.3–25 ms typical (External spec)	P50 (3.79 ms P99) **Measured**	100% chip rejection **By design**	~10 µs wire time **By design**	1–15 ms zero-cross (External spec)

Measured Software Budget (Stages 2–4): `~17.6 ms` min / **`~23 ms` typical** / `~47 ms` P99 • **Full Physical End-to-End (Stages 1–5, estimated — not yet measured on a hardware rig):** `~26 ms` min / **`~48 ms` typical** / `~88 ms` worst-case.

Documented Physical Boundaries: (1) Monocular 2D Projection: Overhead single-camera projects high hand elevation into the 2D Red-Zone and trips conservatively (fail-closed). **(2) Dark-on-Dark Contrast:** Black nitrile gloves over dark cast-iron beds require `≥ 500 lux` LED task lighting. **(3) Spindle Inertia:** Relay cutoff removes drive torque immediately, while heavy bowl blanks coast unless paired with a mechanical/VFD brake; LatheGuard operates alongside physical chip guards and magnetic NVR switches.

4.4 60-Second Judge Reproduction Protocol (Zero Hardware Required)

OFFLINE & CLEAN-ROOM JUDGE VERIFICATION COMMANDS — WINDOWS / LINUX / MACOS

```
# 1. Execute the full 51-test clean-room verification suite (41 LatheGuard + 10 CraftTwin safety math)
python -m pytest services/lathe_guard services/craft_twin/test_safety_math.py -q
# 2. Run the 80-clip synthetic + real studio photo validation benchmark
python services/lathe_guard/corpus/eval_corpus.py
# 3. Launch an 8-second zero-hardware synthetic breach simulation (Offline Tier-0 mode)
python services/lathe_guard/run_lathe_guard.py --mock --visual-breach-after 2 --duration 8 --no-nebius
# 4. Replay recorded workshop breach clip with live bilingual HUD v2 overlay
python services/lathe_guard/run_lathe_guard.py --source services/lathe_guard/corpus/clips/sample_breach_replay.mp4 --mock-serial --preview --duration 10
```

4.5 Conclusion: LatheGuard Physical AI proves that open-weight frontier models on Nebius Token Factory and deterministic safety engineering can protect human hands and preserve heritage handcraft. Honoring the memory and workshop discipline of **Master Mostafa Fahmy**, it ensures that self-healing craft sanctuaries can continue to spin safely for generations to come.